

Homework 4

Solutions must be **typed** and **submitted electronically**.

1. Compute $\text{BirthdayProb}(50, 512)$ and $\text{BirthdayProb}(5, 512)$ exactly, to 4 significant digits. Compare to the upper and lower bounds proven in Lemma 4.10.
2. Prove the following generalization of the results in this chapter:

Fix a value $x \in \{0, 1\}^\lambda$. Then when taking q uniform samples from $\{0, 1\}^\lambda$, the probability that there exist two distinct samples **whose XOR is x** is $\text{BirthdayProb}(q, 2^\lambda)$.

3. Mining Bitcoin requires evaluating the SHA-256 cryptographic hash function as many times as possible, as fast as possible. When mining Bitcoin today, the only economically rational choice is to use special-purpose hardware that does nothing except evaluate SHA-256, but is millions (maybe billions) of times more energy efficient than a general-purpose CPU evaluating SHA-256.
 - (a) The relevant specs for Bitcoin mining hardware are wattage and giga-hashes (or tera-hashes) per second, which can be converted into raw energy required per hash. Search online and find the most energy efficient mining hardware you can (e.g., least joules per hash).
 - (b) Find the cheapest real-world electricity rates you can, anywhere in the world. Use these to estimate the monetary cost of computing $2^{40}, 2^{50}, \dots, 2^{120}$ SHA-256 hashes.